Reasoning Over Fully-occluded Obstacle for Visual Non-prehensile Loco-manipulation

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Introduction

Our visual non-prehensile manipulation policies use onboard cameras to move large objects without relying on external object-tracking systems. During manipulation, however, the object may fully occlude obstacles, leading to unexpected contacts that alter robot–object dynamics. Our system detects deviations from nominal dynamics predictions to identify these hidden interactions, then coordinates active perception, obstacle removal, and task resumption.

Input-output

Input: Three-frame segmented-depth histories from gripper, front-left, and front-right cameras, robot proprioception, previous action, and a local object-pose subgoal.

Output: 12 commands: 6-DoF base velocity and six arm-joint commands.

Framework

Our framework has three layers. (1) The first layer uses a dynamics-aware task planner to generate subgoals and monitor transitions.
(2) The second layer uses a distilled visual policy to map onboard observations and subgoals to action commands.
(3) The third layer uses a low-level loco-manipulation controller to execute these commands.
Detected anomalies trigger active perception, obstacle removal, and task resumption.

Experiment

Loco-Manipulation: Teacher and visual-student policies were tested on chair and cuboid tasks over 256 episodes per condition. Goals varied in position and orientation, and success required errors below 0.15 m and 15°. Evaluation compared goal-hit rate and terminal position and rotation errors.

Detection: Separate evaluation episodes measured episode-level nominal FPR, post-contact TPR, and median delay among successful post-contact detections. Our detector achieves 80.2% post-contact TPR with 8.0% nominal FPR and a 1.64 s median delay among successful post-contact detections, providing the strongest balance among the nonprivileged detectors.

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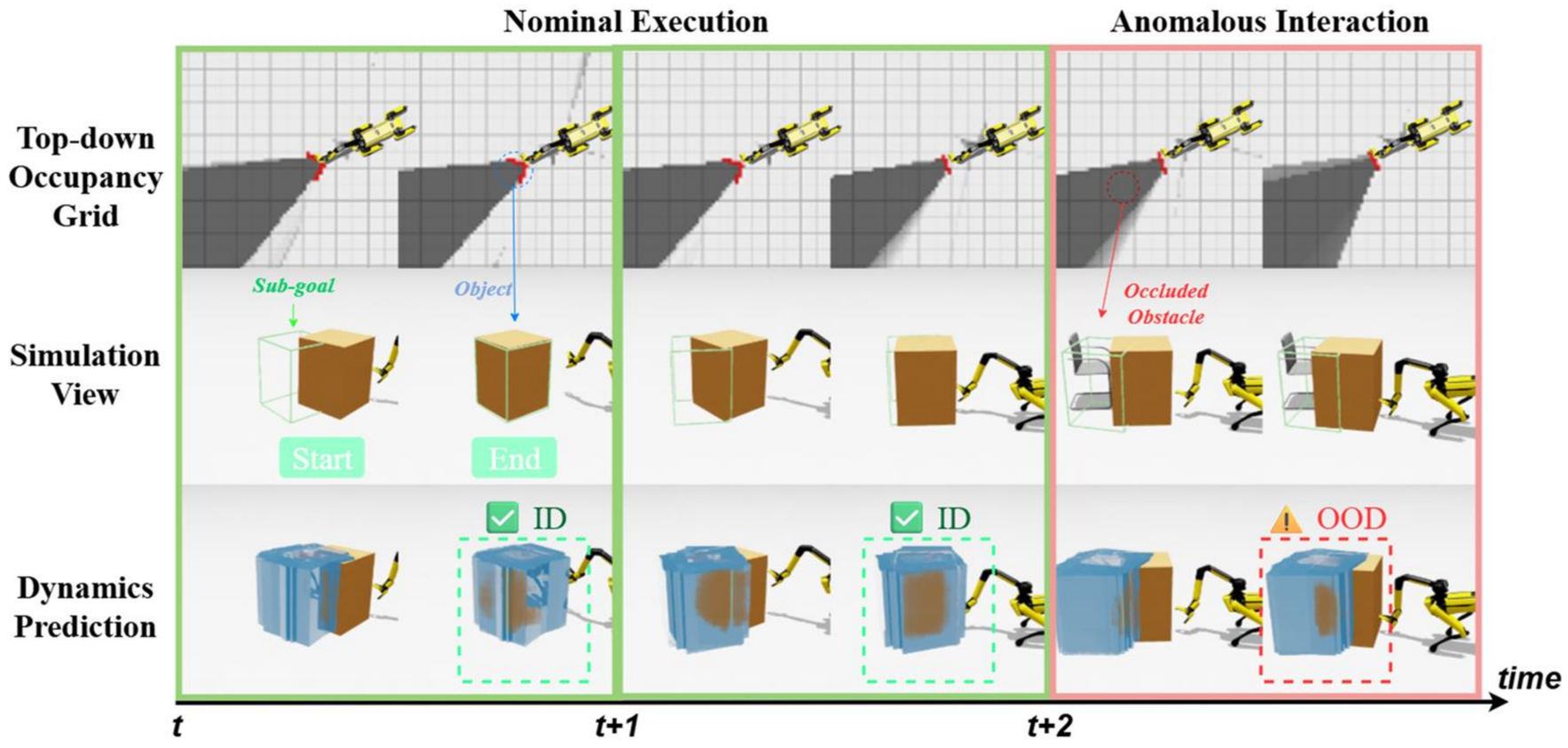
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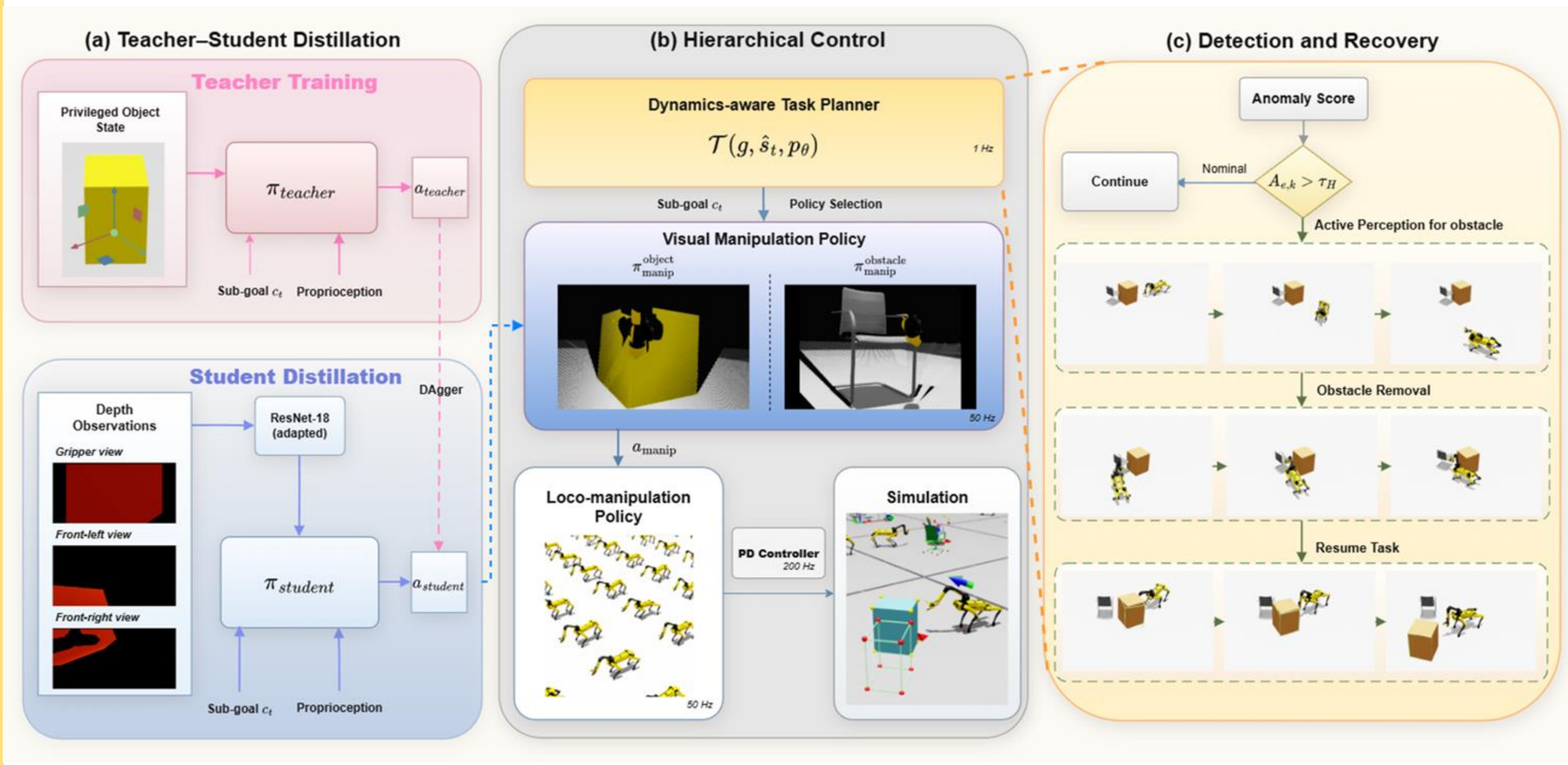
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Dynamics prediction enables reasoning about fully occluded obstacles during visual loco-manipulation

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Framework Overview



Detection Evaluation

Detector	Nominal FPR (%) ↓	Early alarm (%) ↓	Post-contact TPR (%) ↑	Miss (%) ↓	Median delay (s) ↓
Object stall	0.0	0.0	3.5	96.5	4.62
Object L2 (0.1 m)	79.0	50.2	49.6	0.2	0.58
Object L2 (0.5 m)	2.0	0.0	10.5	89.5	3.82
Energy score (ours)	**8.0**	**2.1**	**80.2**	**17.7**	**1.64**
Contact oracle*	0.0	0.0	100.0	0.0	0.49

* The contact oracle uses privileged contact information. Early alarm + post-contact TPR + miss = 100%. Delay is the median among successful post-contact detections.